



SYNTECH

Empowering Tomorrow's Technology

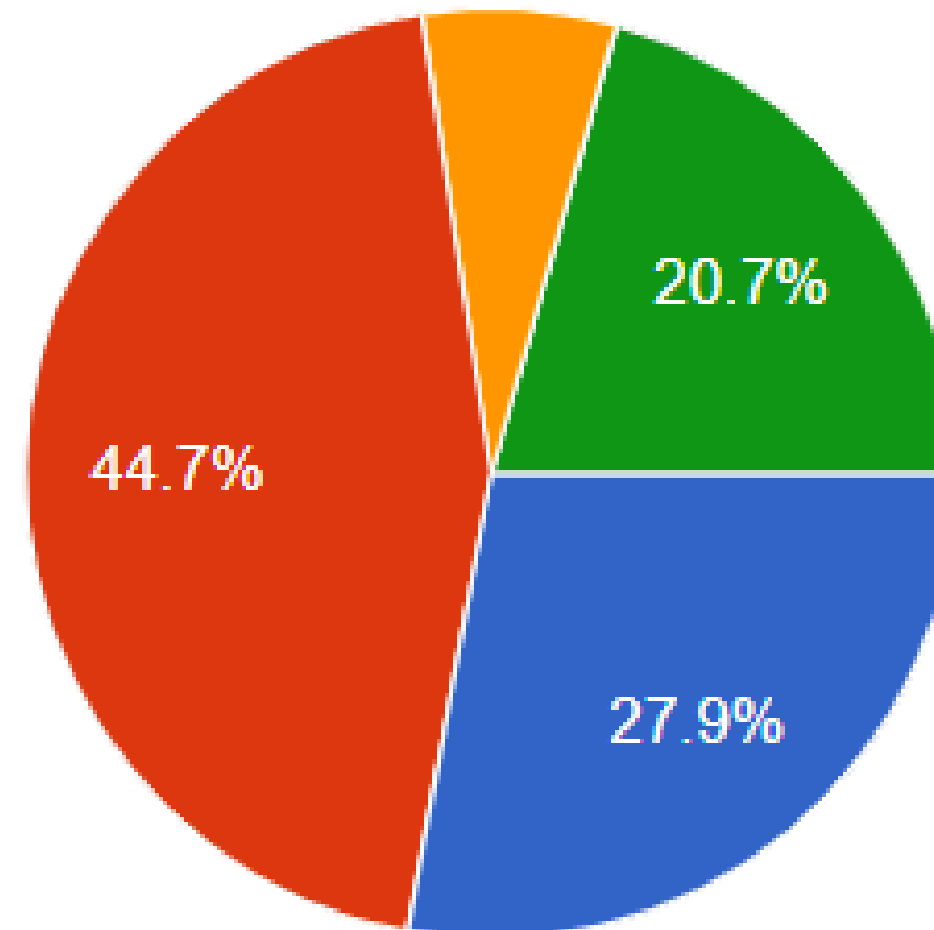




**Are you satisfied with
the current bus
transportation?**

How often do you use public bus system?

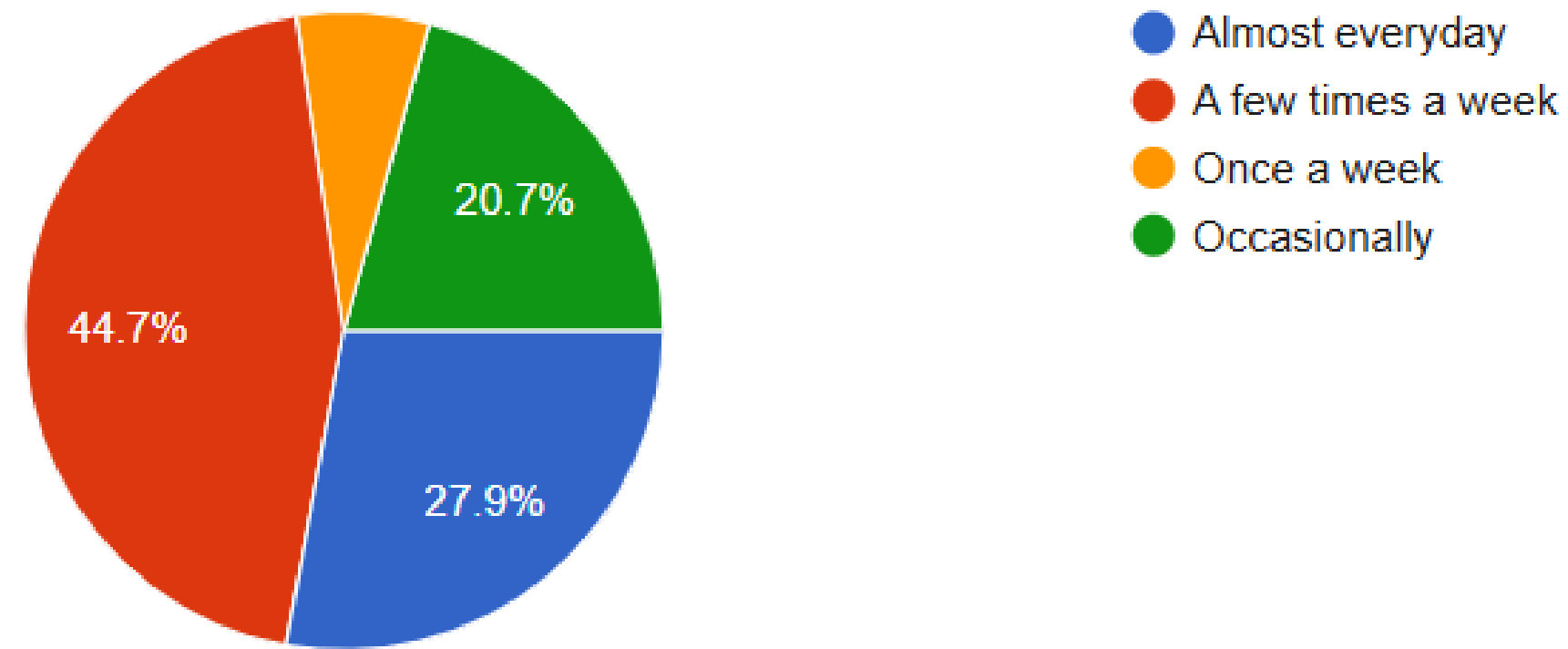
208 responses



- Almost everyday
- A few times a week
- Once a week
- Occasionally

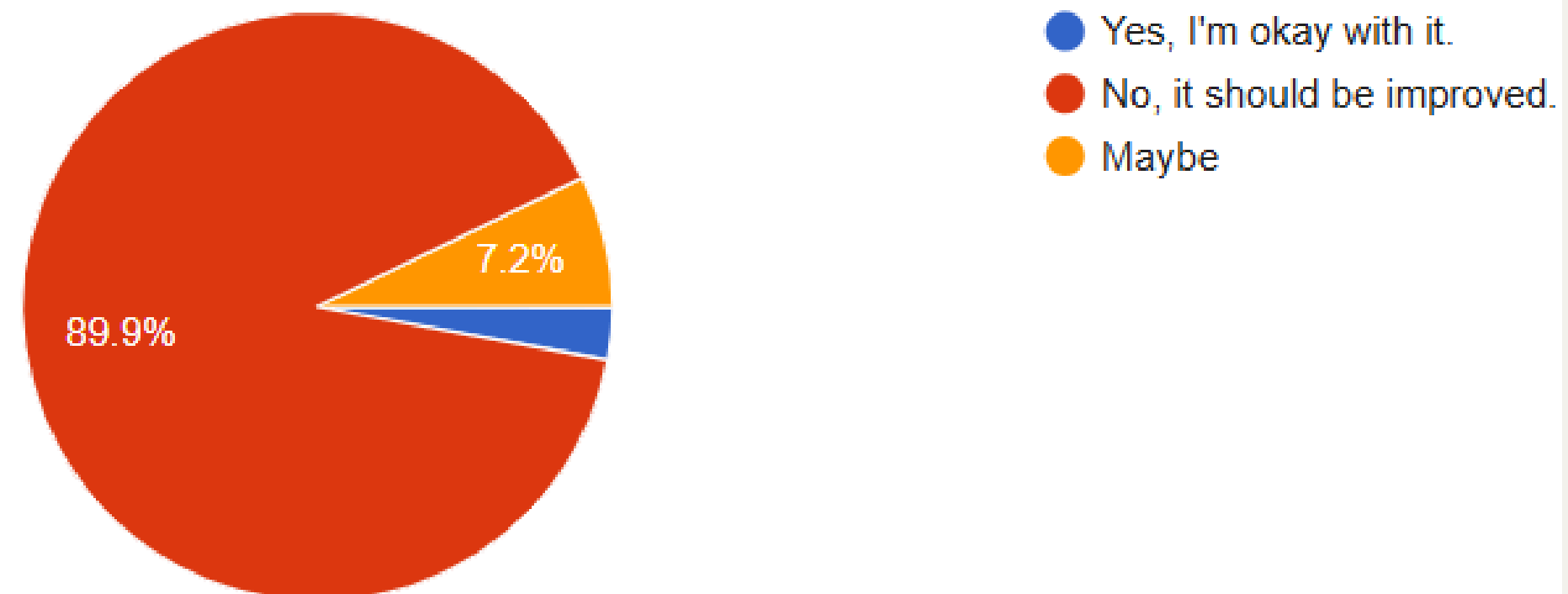
How often do you use public bus system?

208 responses



Are you satisfied with the current public bus transportation systems?

208 responses





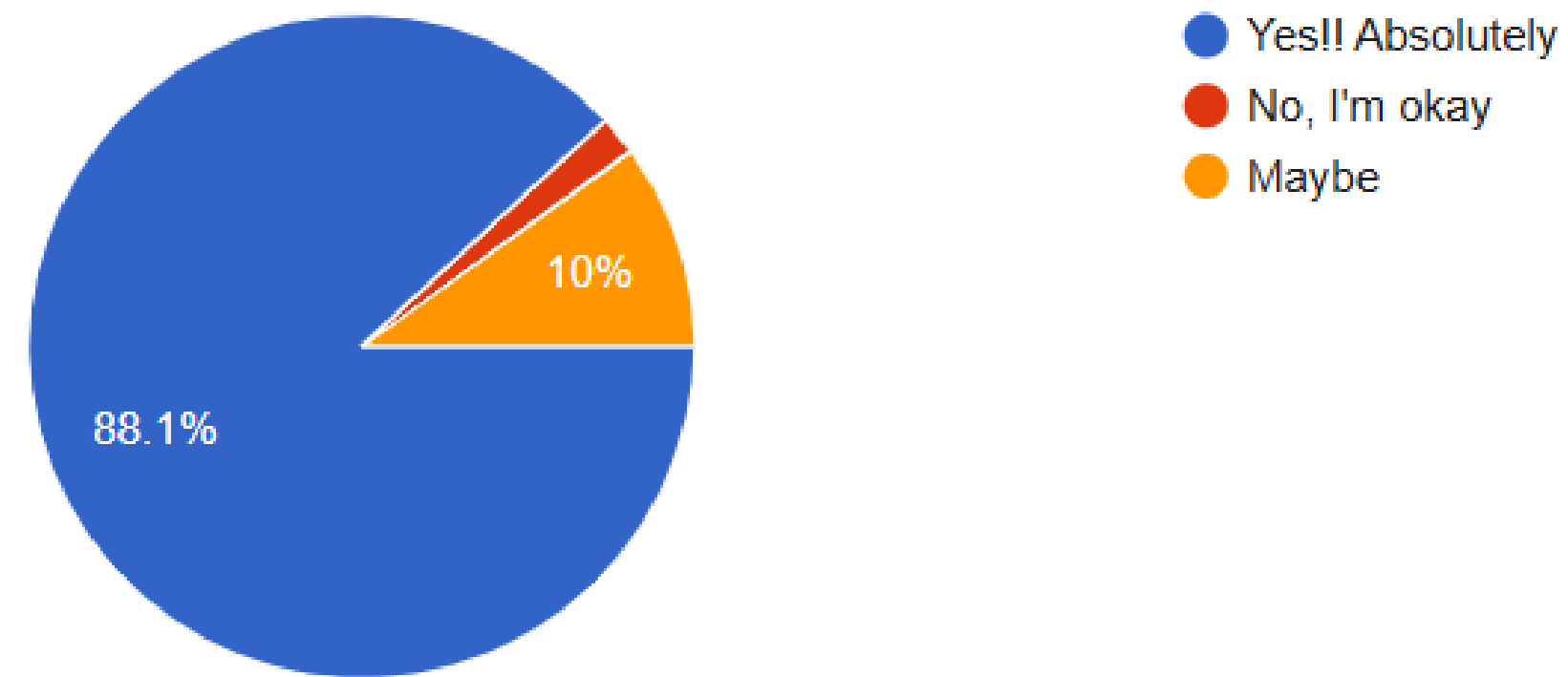
Problem Statement

We have observed and identified these challenges faced by almost everyone using public bus transportation

- Overcrowdiness
- Not properly running on a time schedule
- Irresponsible / dangerous driving patterns
- Not enough change money to pay
- Not being able to complain bus misconducts
- No idea about the correct bus route

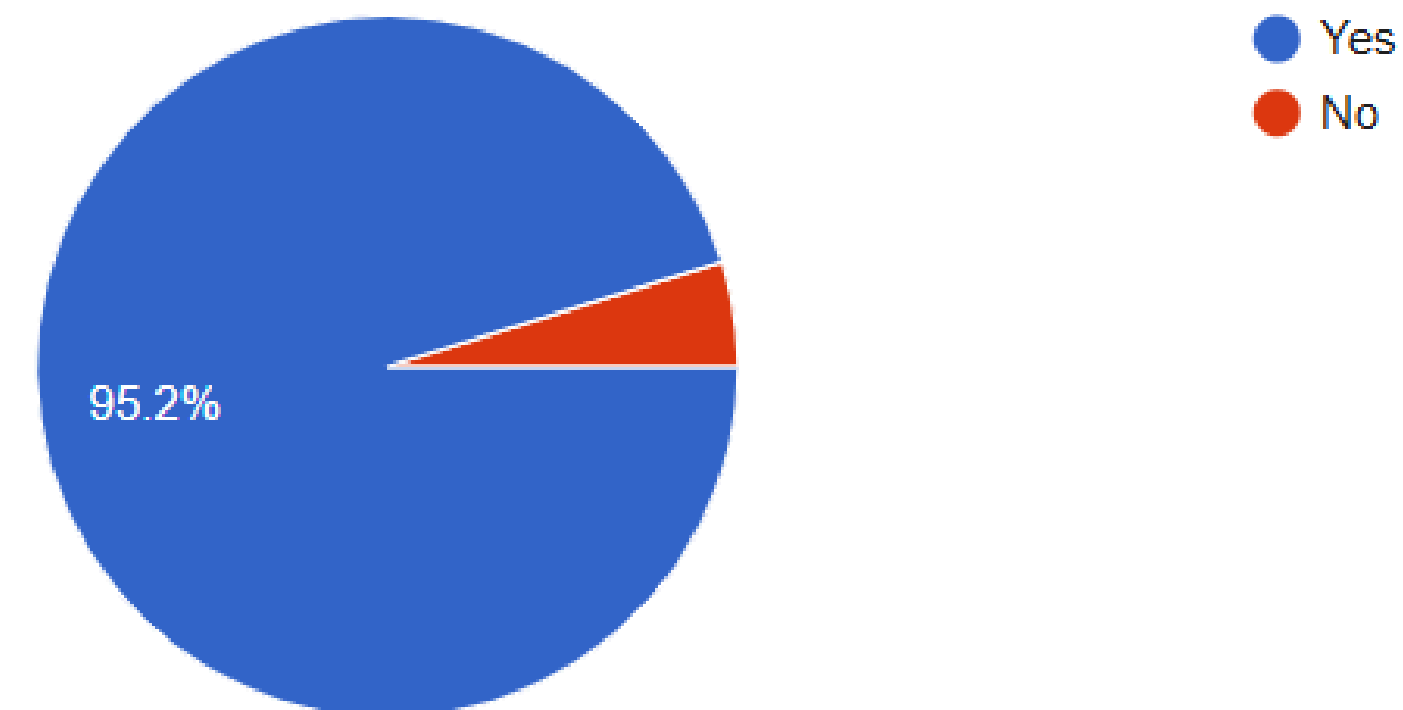
Would you like to monitor the buses and bus routes near you via an app?

210 responses



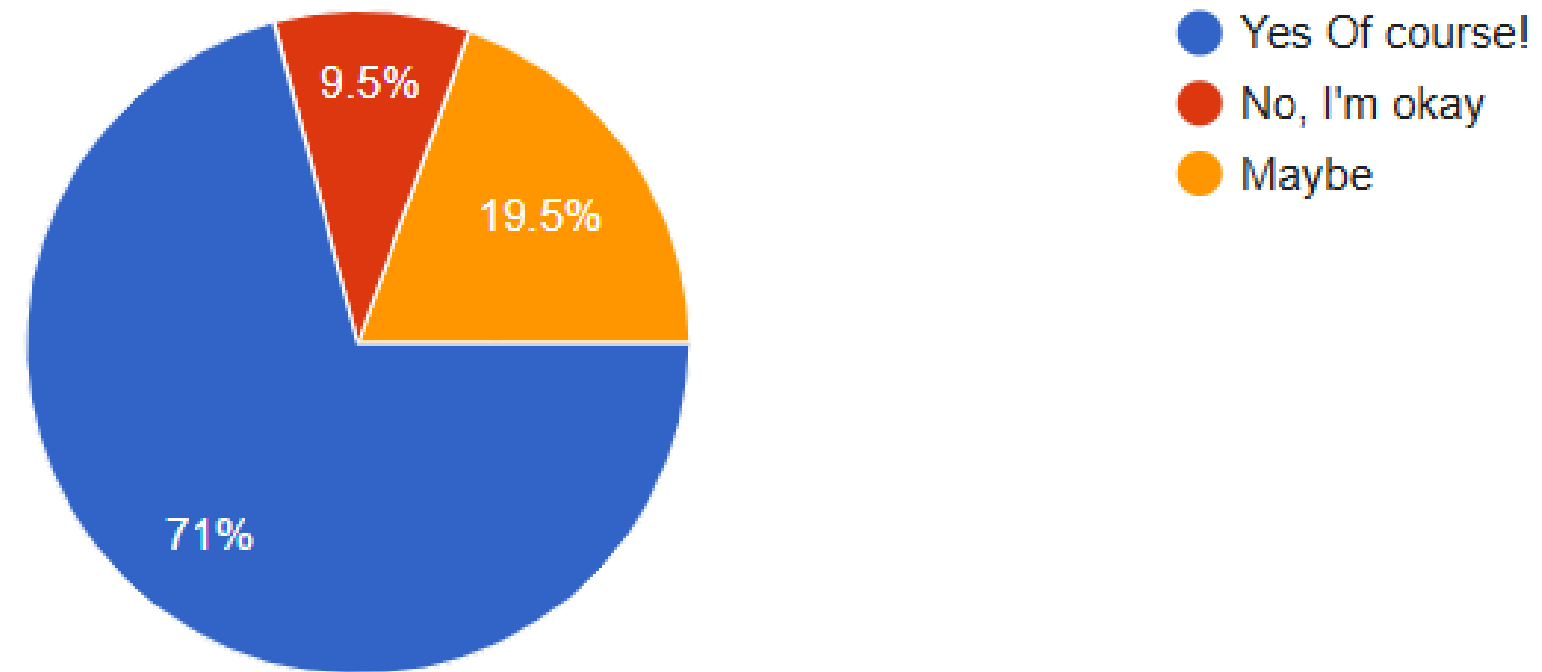
Would it be convenient to know the estimated arrival time of the next bus to your nearest bus halt?

207 responses



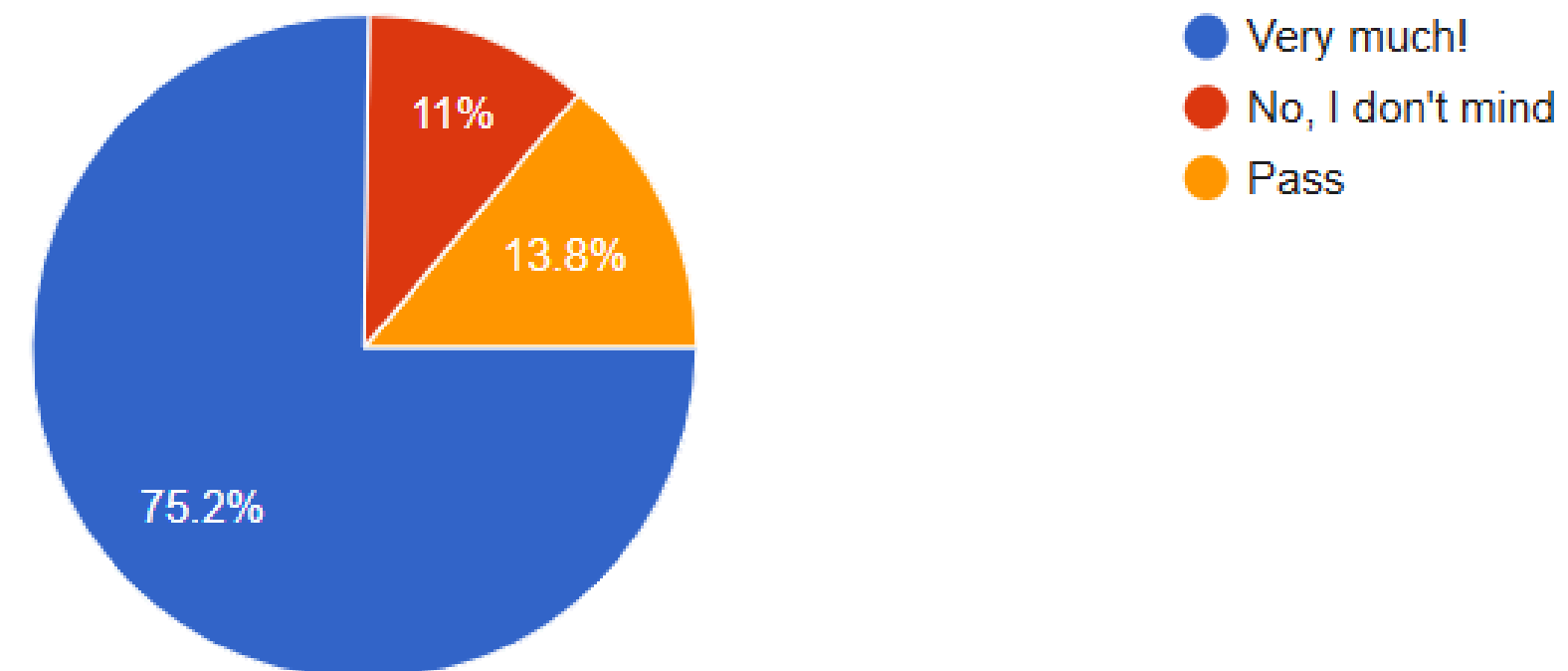
Would it be helpful if you could book bus tickets through an app?

210 responses



How convenient would it be if you could make complaints about bus misconducts via the app?

210 responses





What Can We Do

To Regulate And Monitor Busses

A fixed GPS tracker attached to the bus.

A community based data collection system

Intergrate a GPS tracker and a bus data monitoring unit to get insights

Using legal officers to monitor busses constantly



An IoT Enabled Ticketing Machine

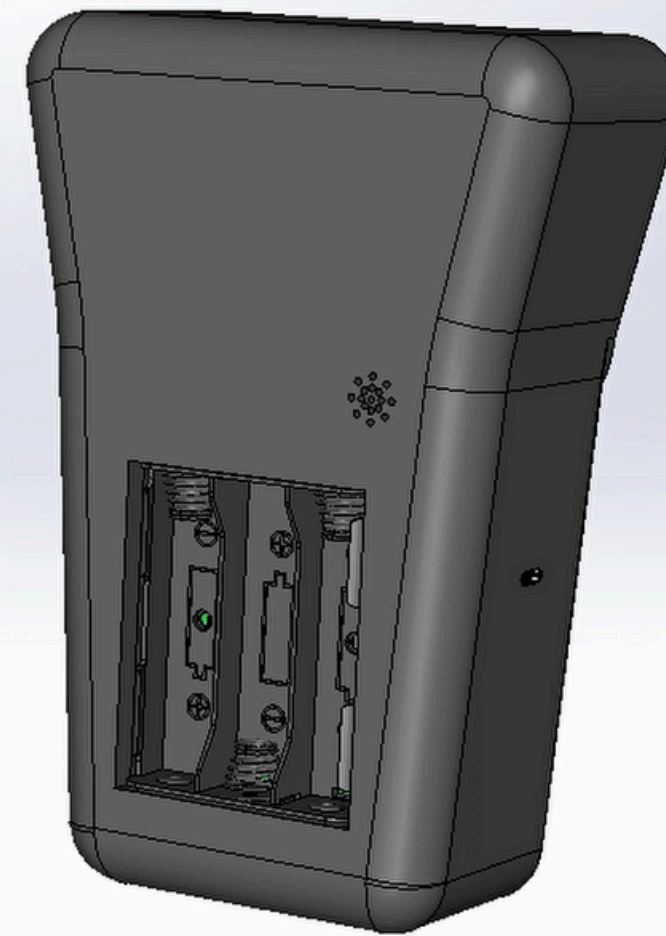
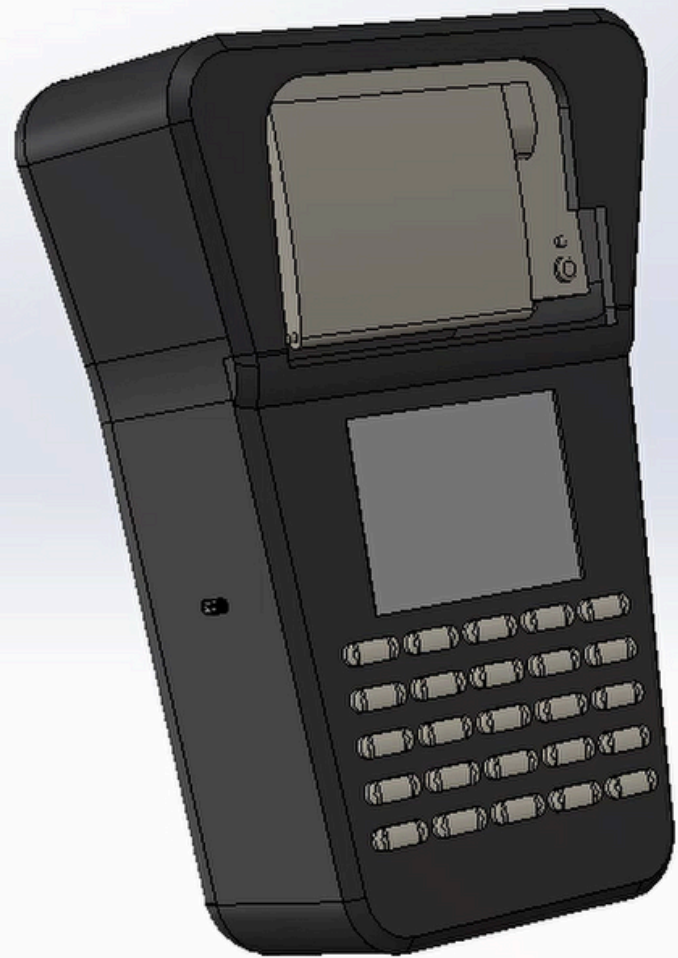
A ticketing machine similar to the existing bus ticketing machine

Intergrated GPS tracker

Real time motion detection

Passesnger count feed

IoT enabled Bus Ticketing Device



Device Features

Our Device act as a ordinary bus ticketing machine from the bus conductors end but with IoT capabilities

- TFT LCD display with 5 x 5 scanning Keyboard



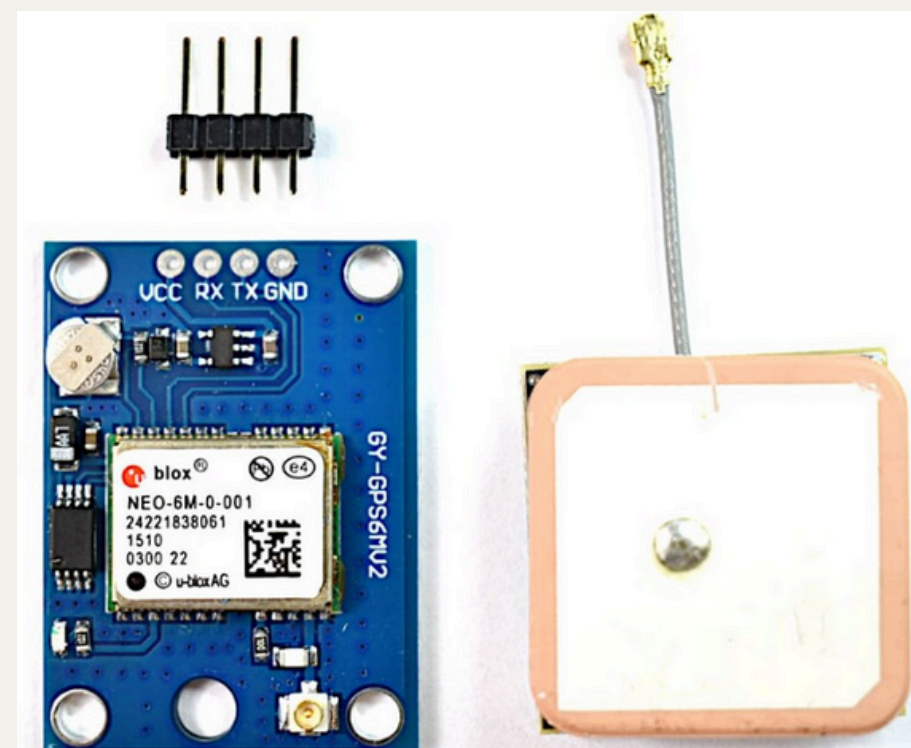
- 58mm Thermal Printer



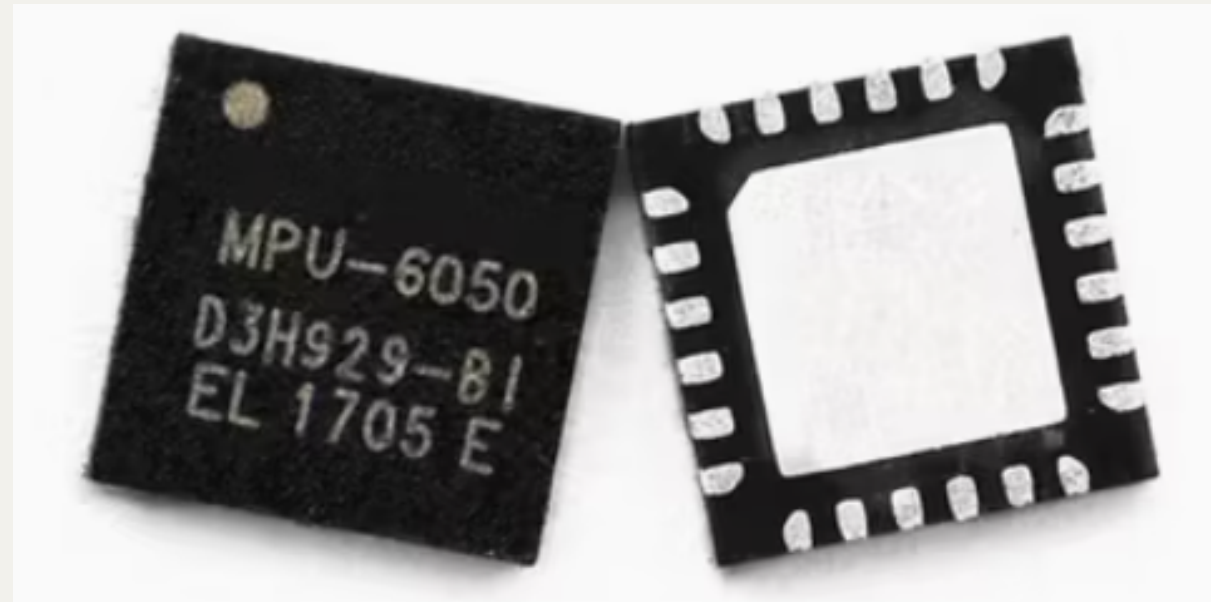
- GPS module for acquiring GPS data



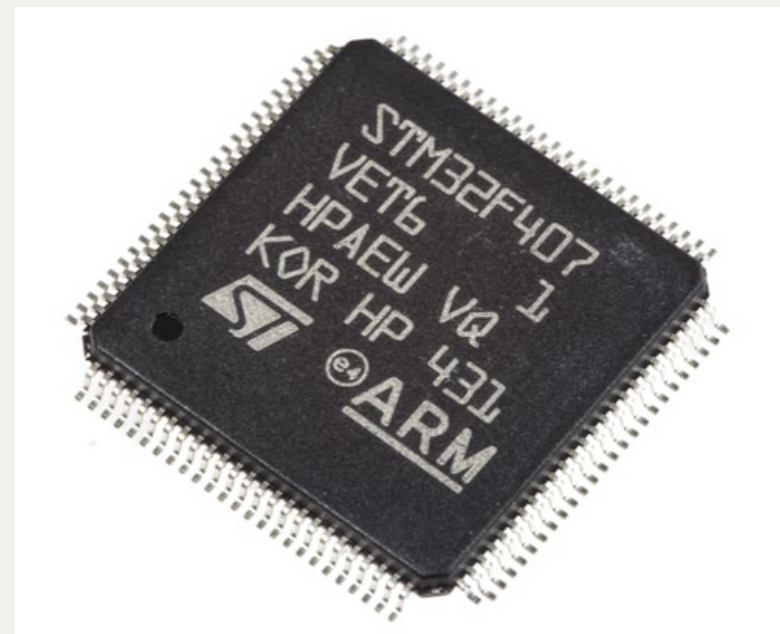
- GPRS module for cellular network connection

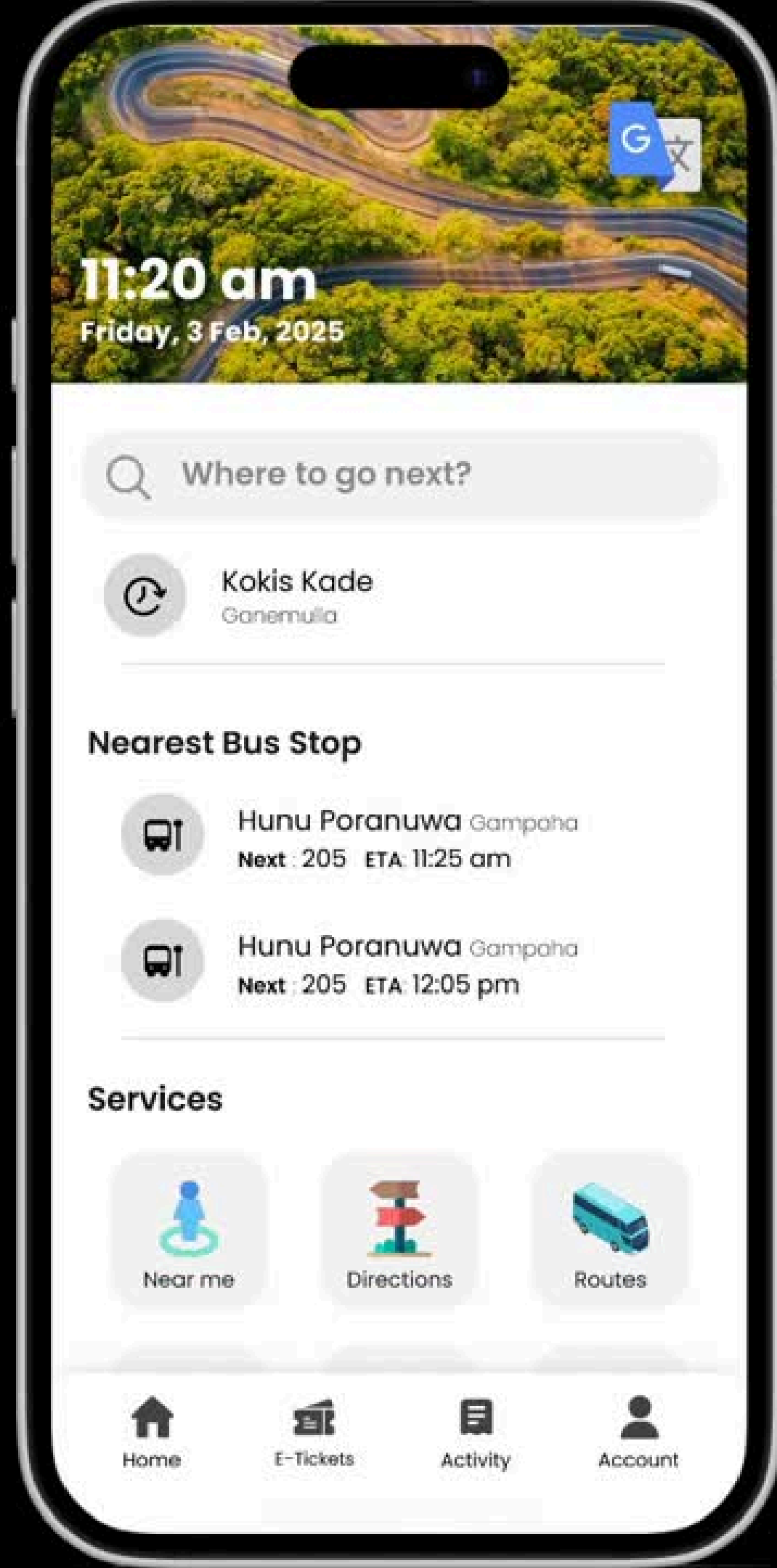


- IMU module for detecting irregularities and emergencies



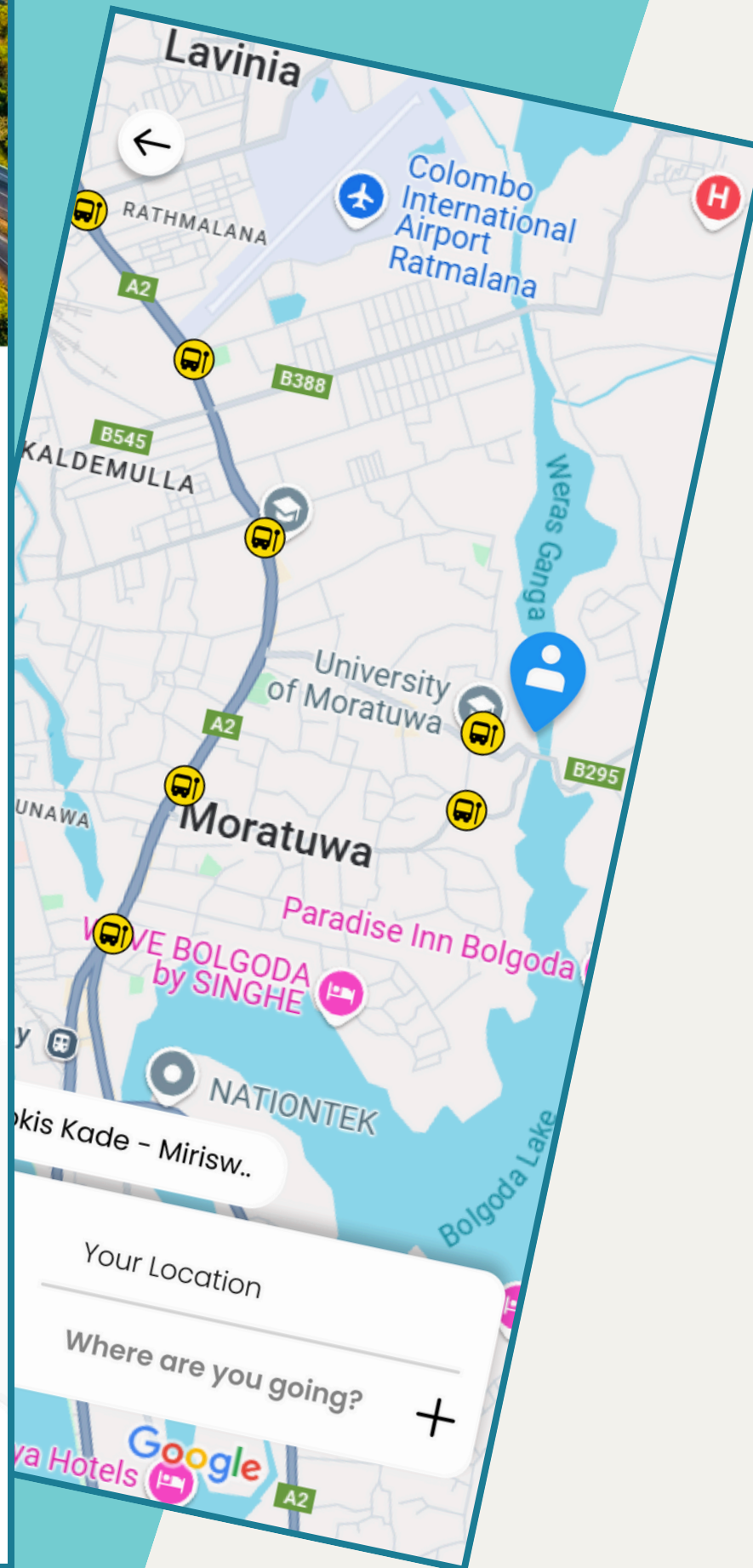
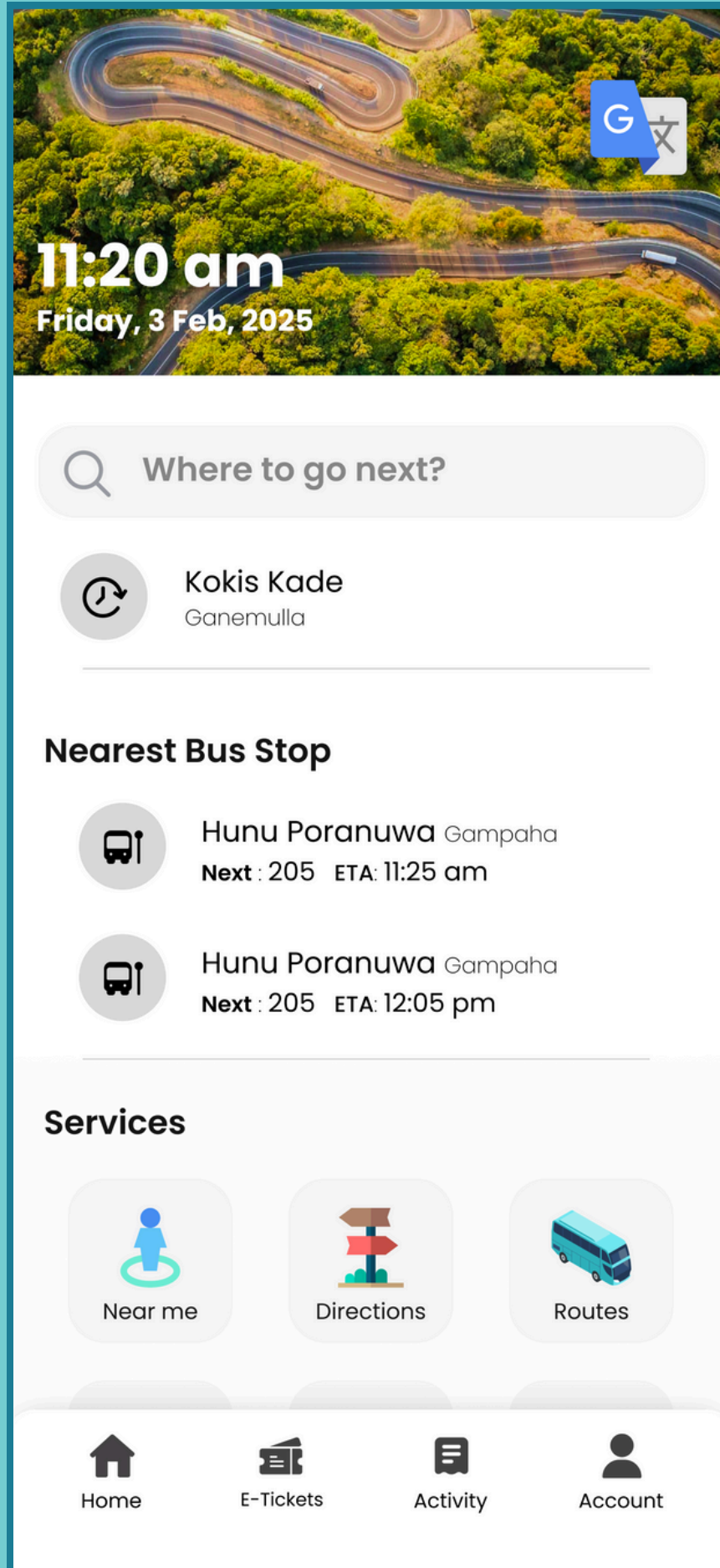
- STM32F4 for low power operation





SYNTRACK

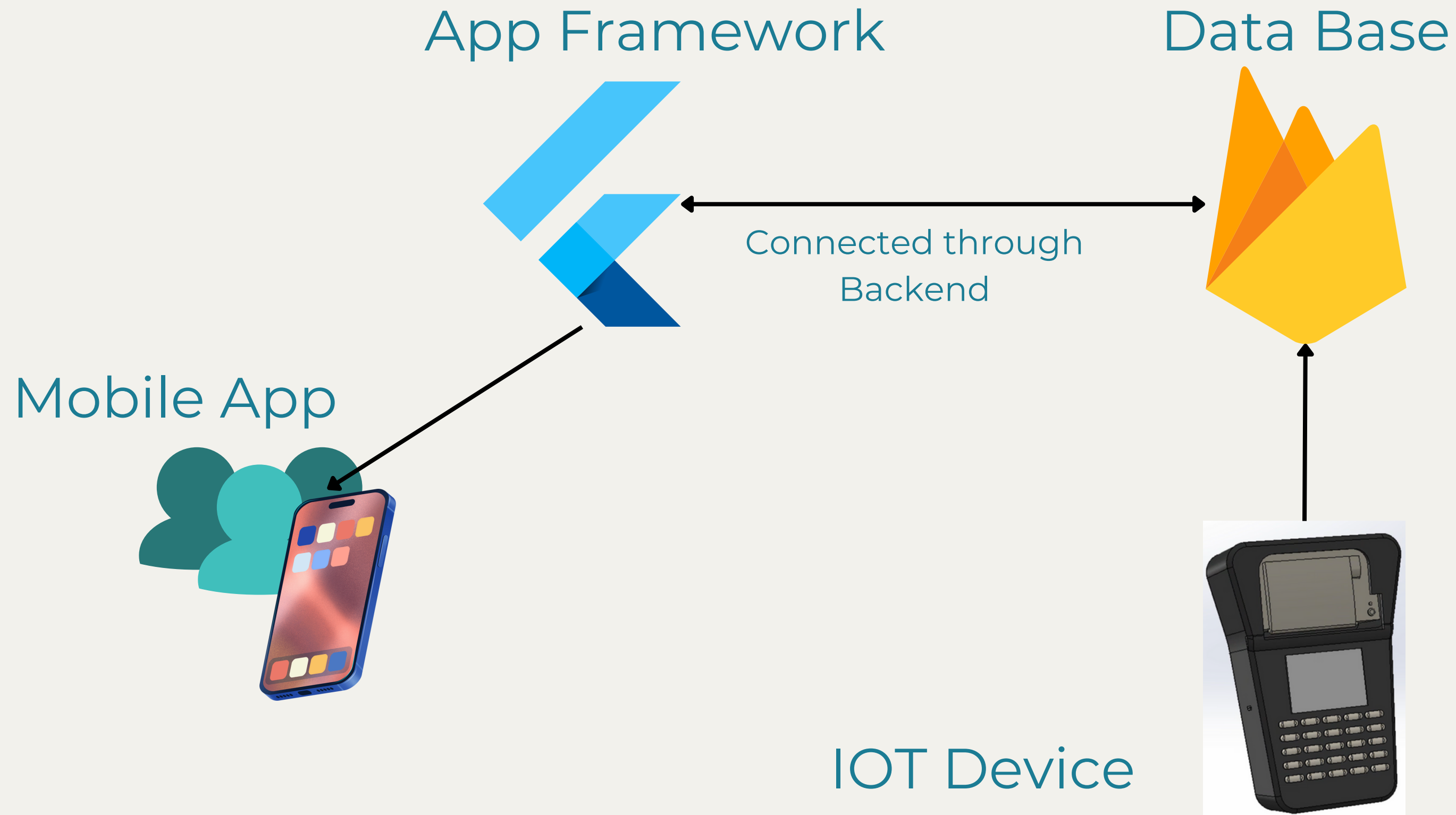
Commuter App



DESIGN

- Familiar Designs
- Accessibility

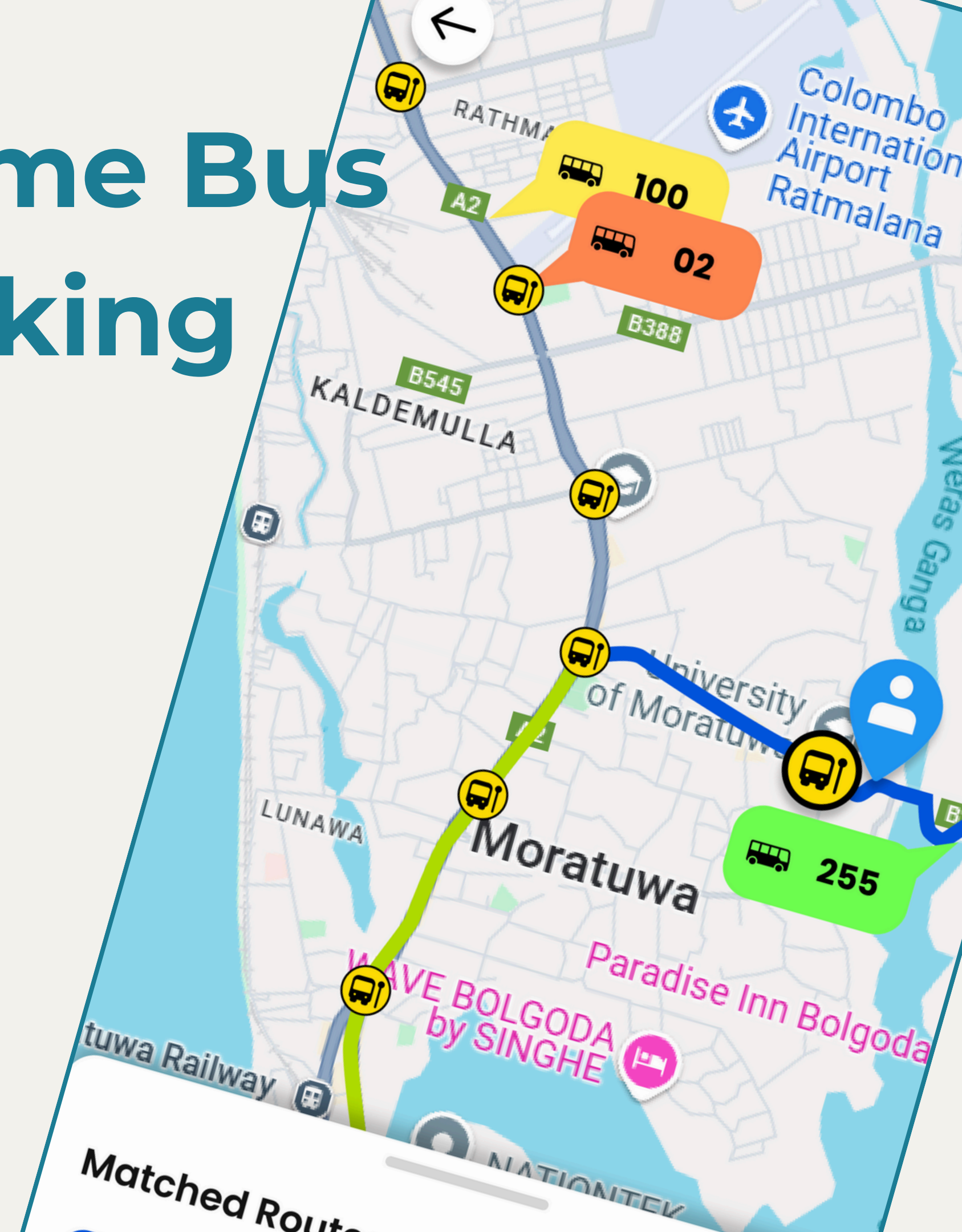
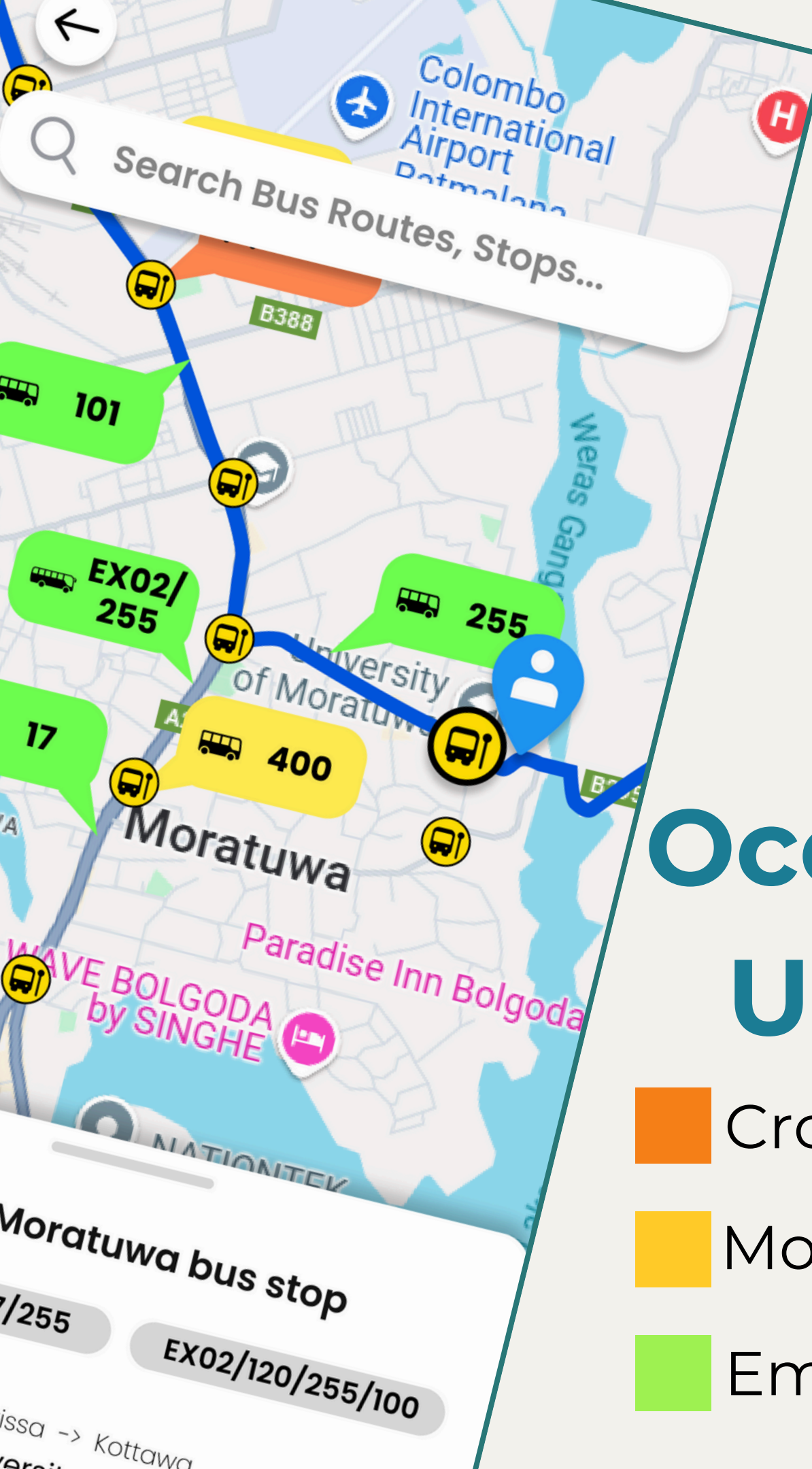
IMPLEMENTATION

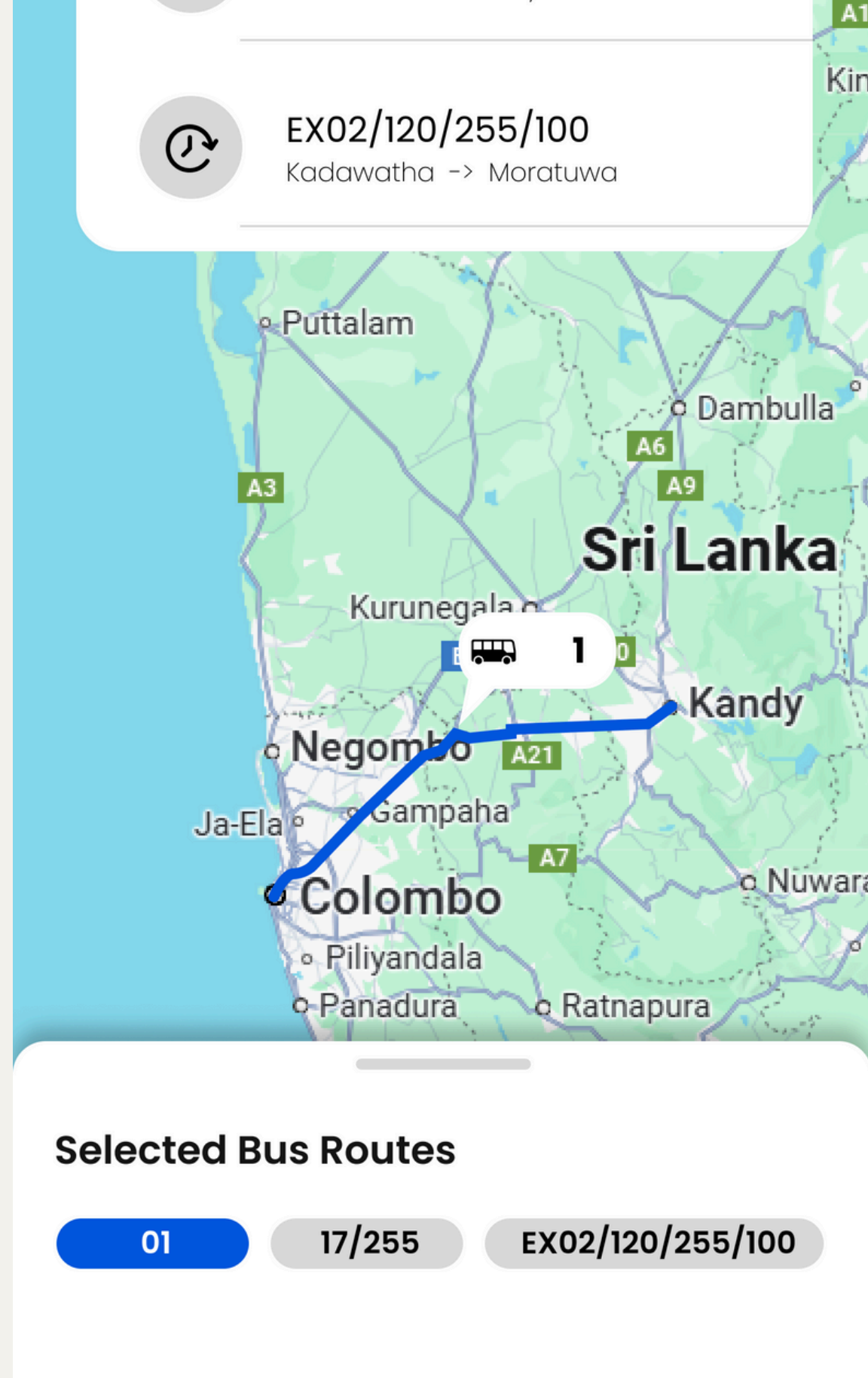


Real Time Bus Tracking

Occupancy Updates

-  Crowded
-  Moderately Crowded
-  Empty





Route Info

Commuter Support

Emergency Contact

In case of an emergency, directly contact the authorities

Your Location

6°47'40.4"N 79°54'08.6"E

Police Emergency Hotline (119)

Police Emergency Hotline (11)

Ambulance / Fire & Rescur

Accident Service Hospit

mb Disposal

Create New Report

Category

Priority

Description

Location

My Location

Vehicle (Bus) Registration Number

Contact Details

☐ Remain Anonymous

Name

Email

Ticketing

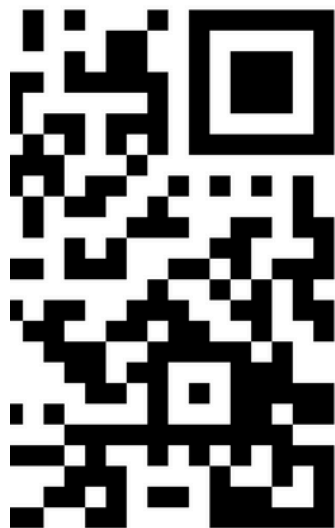
Available Tickets



Katubadde Moratuwa
Kdawatha Kadawatha
valid till: 11:25 am 02/04/2025



idde Moratuwa
ltha Kadawatha
11:25 am 02/04/2025



Purchased Tickets

ts

No Tickets Found

Found

Notifications

Road Accidents have
been reduced by 20%
this year



All Bus Fees has been
reduced by 5%



Read more



Road Safety in Sri Lanka
A Growing Concern



How to get around Sri Lanka
Travellers Isle



Transport Development
Ceylon, 1800-1947

Road Accidents have
been reduced by 20%
this year

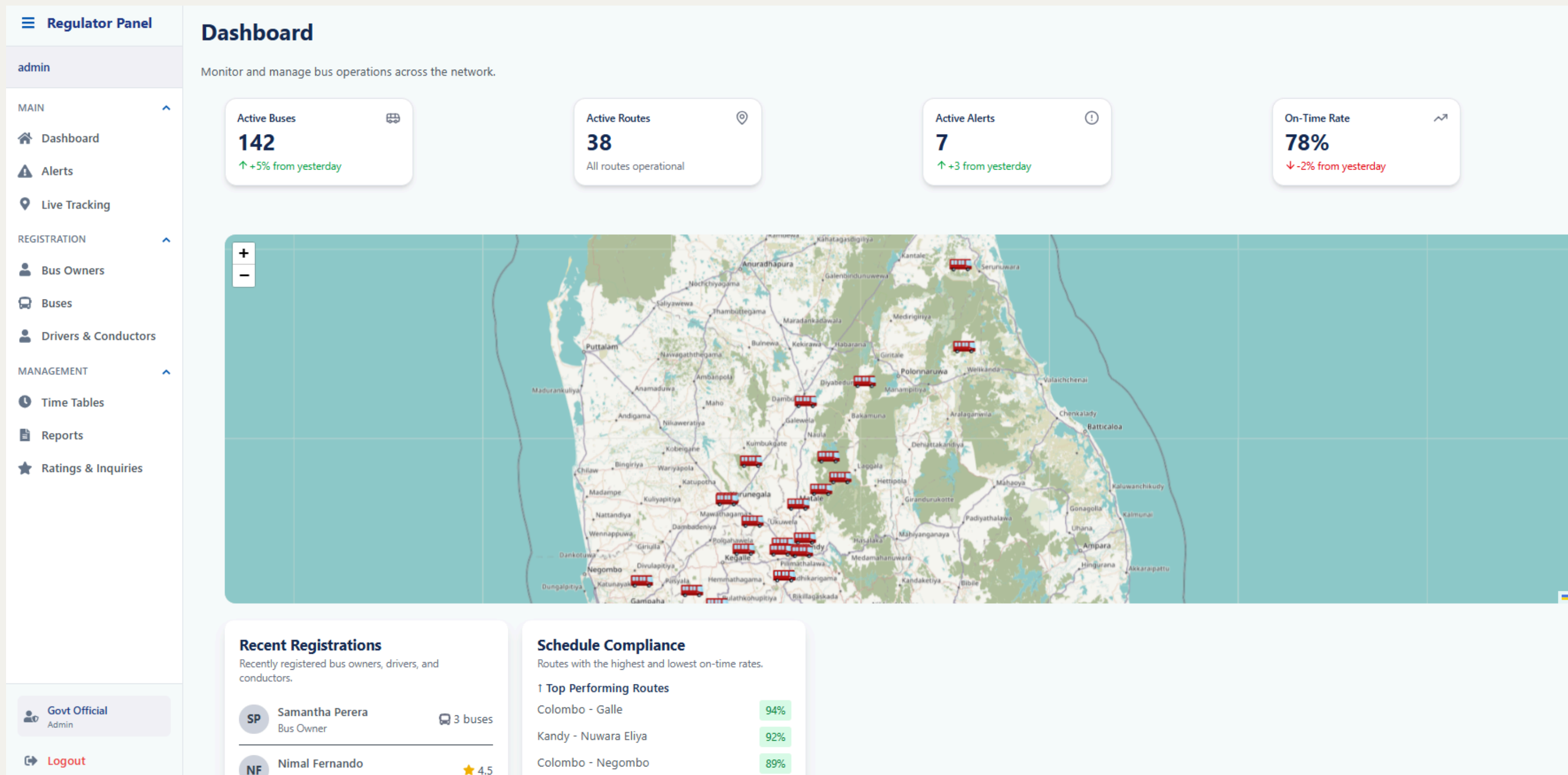


SYNTRACK

Commuter App

Smoother Safer Predictable

Web Platform





Web Platform

- Live-tracking of all the busses in the network
- Bus owner and operator registration

- Automated alerts for irresponsible and dangerous driving behavior
- Quarterly report generation featuring optimized resource allocation recommendations based on analyzed commuter pattern data.



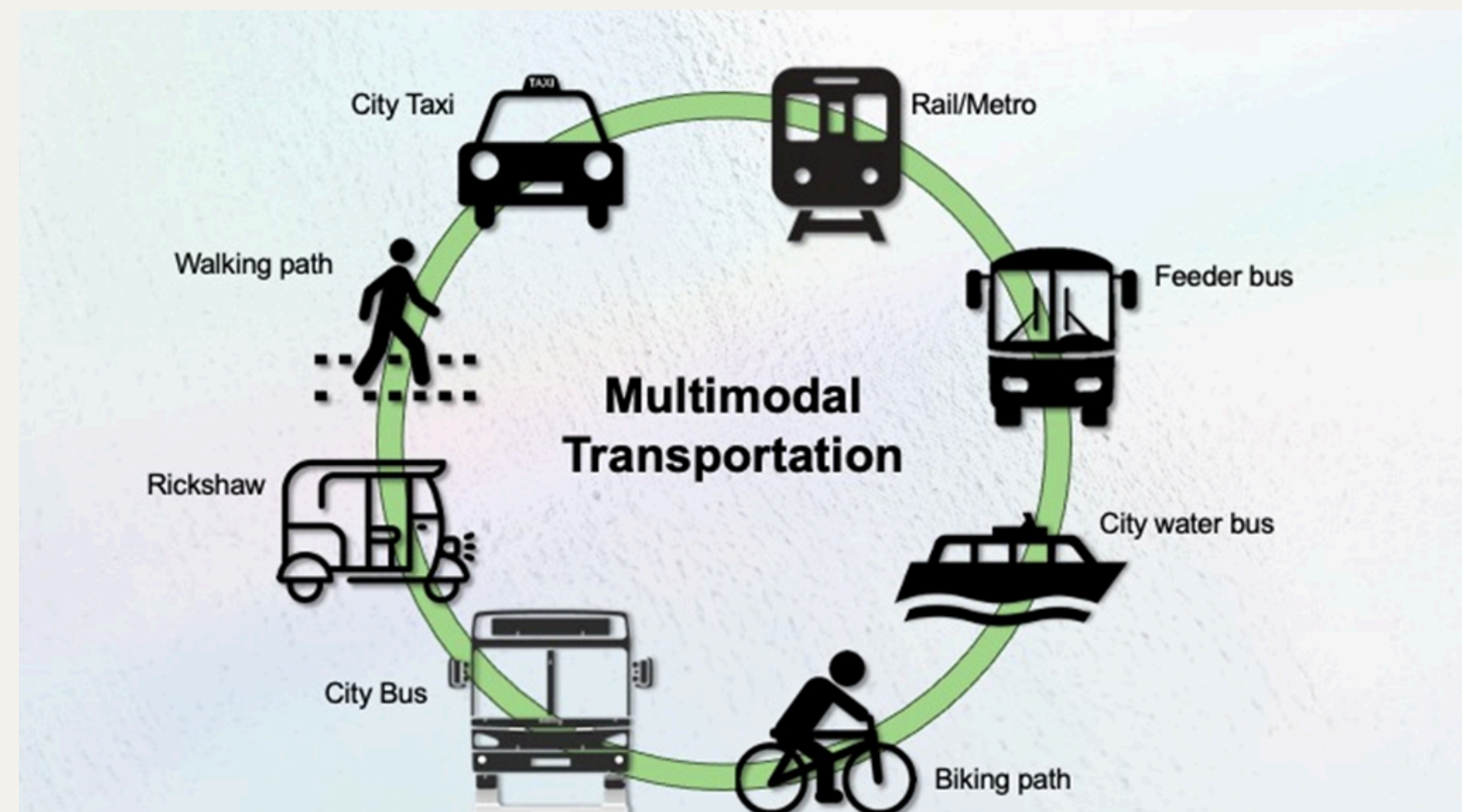
Future Development

- Adaptive resource allocation for routes using collected commuter patterns

- Aiming for smart city concepts that aligns with Sustainable Development Goals (SDG 17)

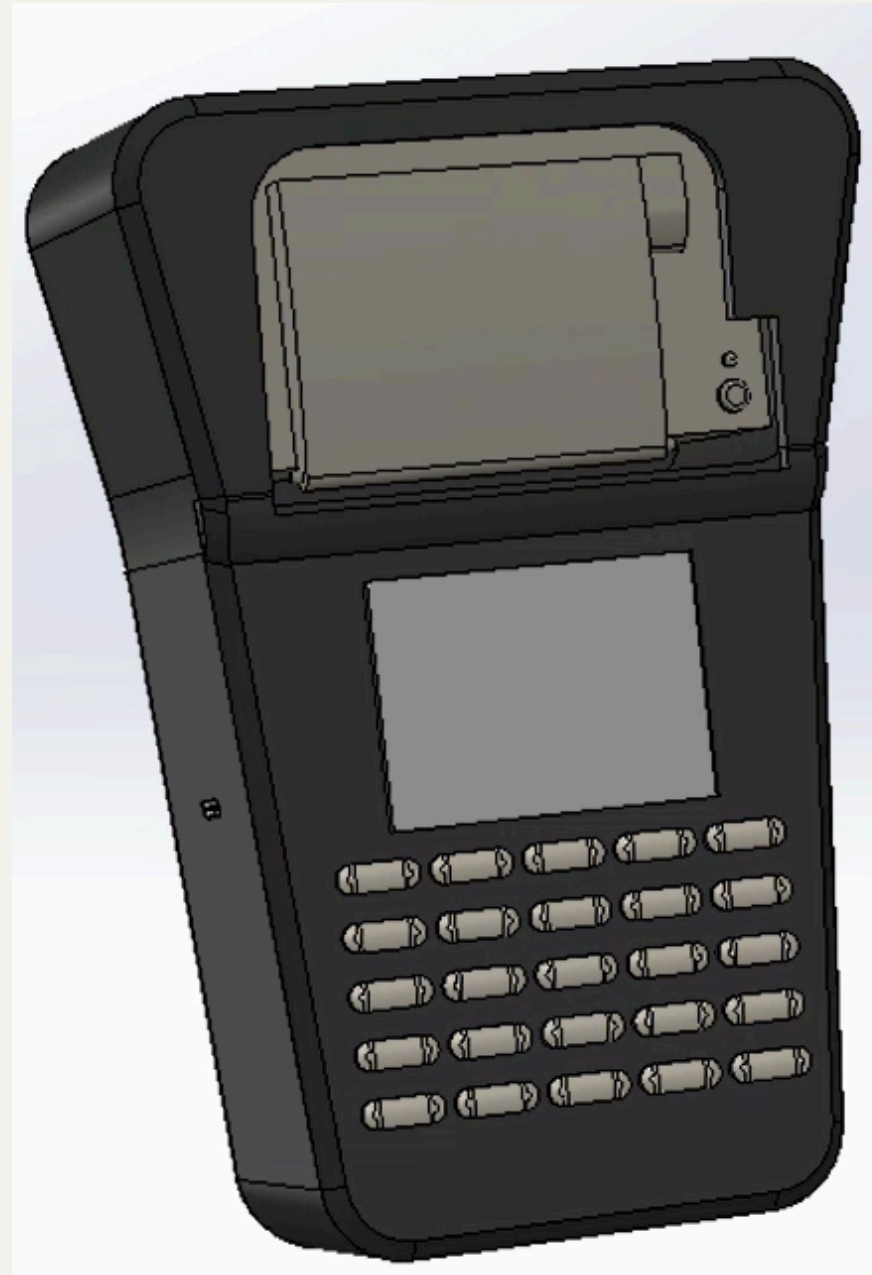


- Integrate multiple transportation modes and collaborate with service providers to deploy a unified platform that ensures seamless, affordable, and comfortable commuting across the entire island.

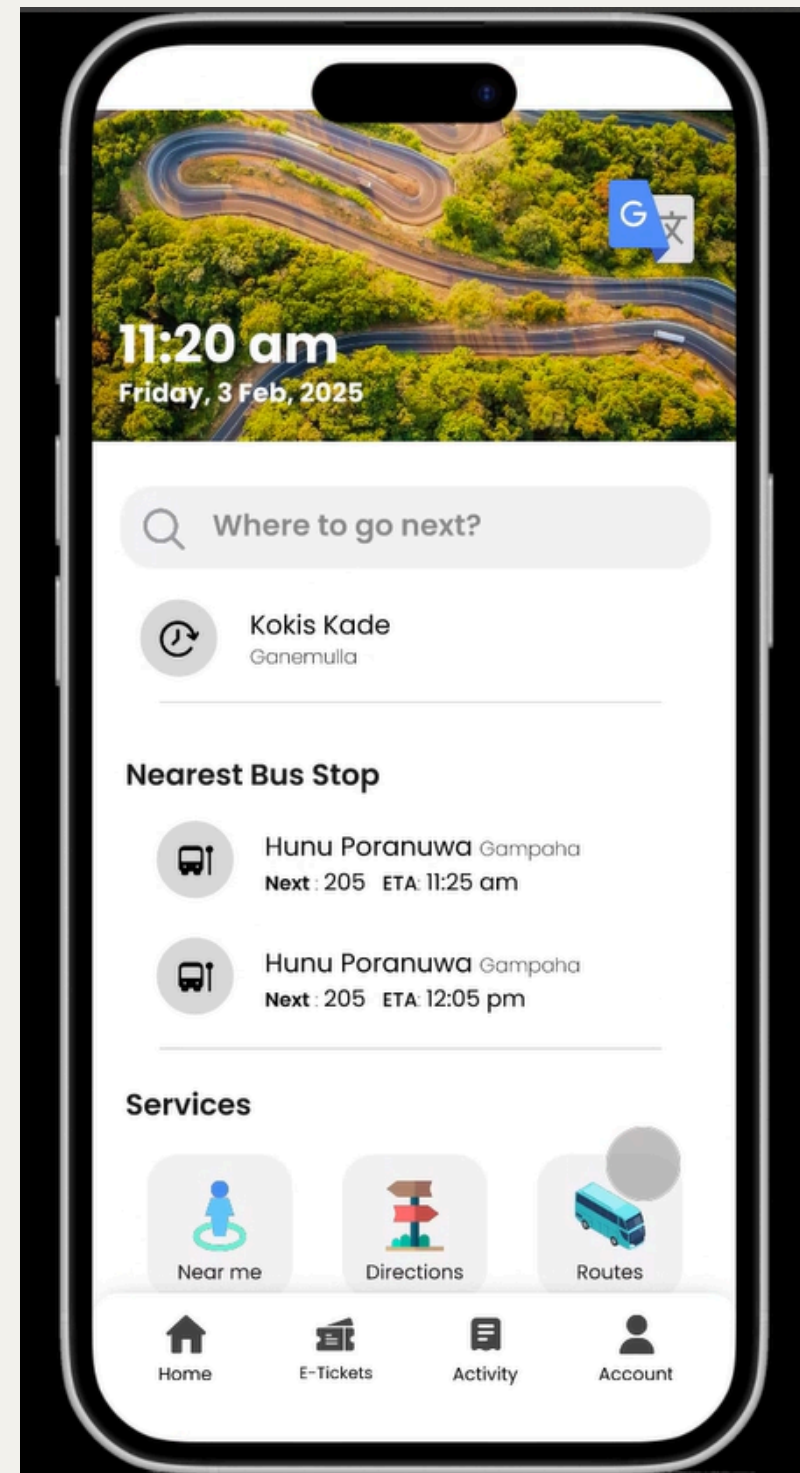


Our progress

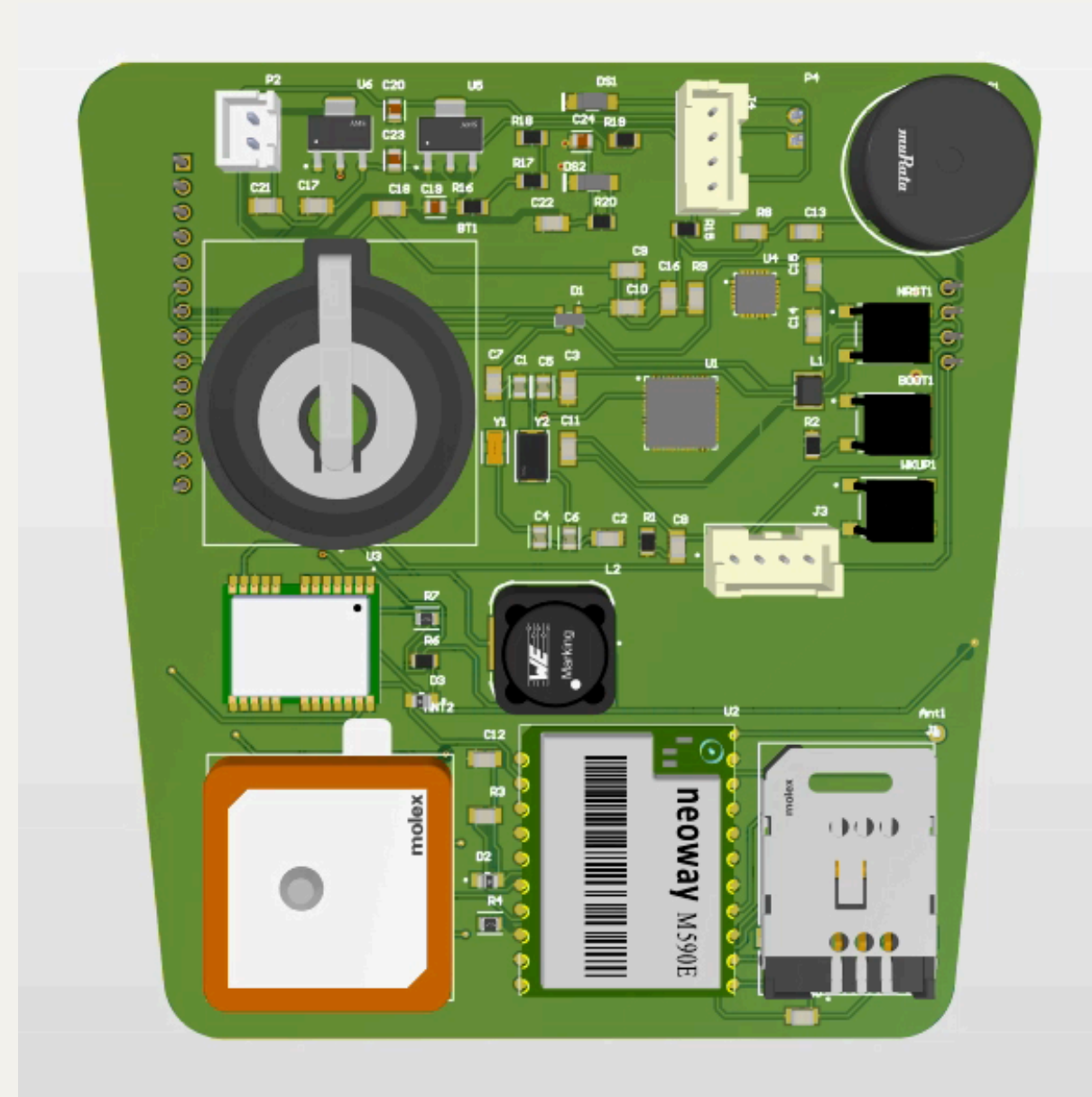
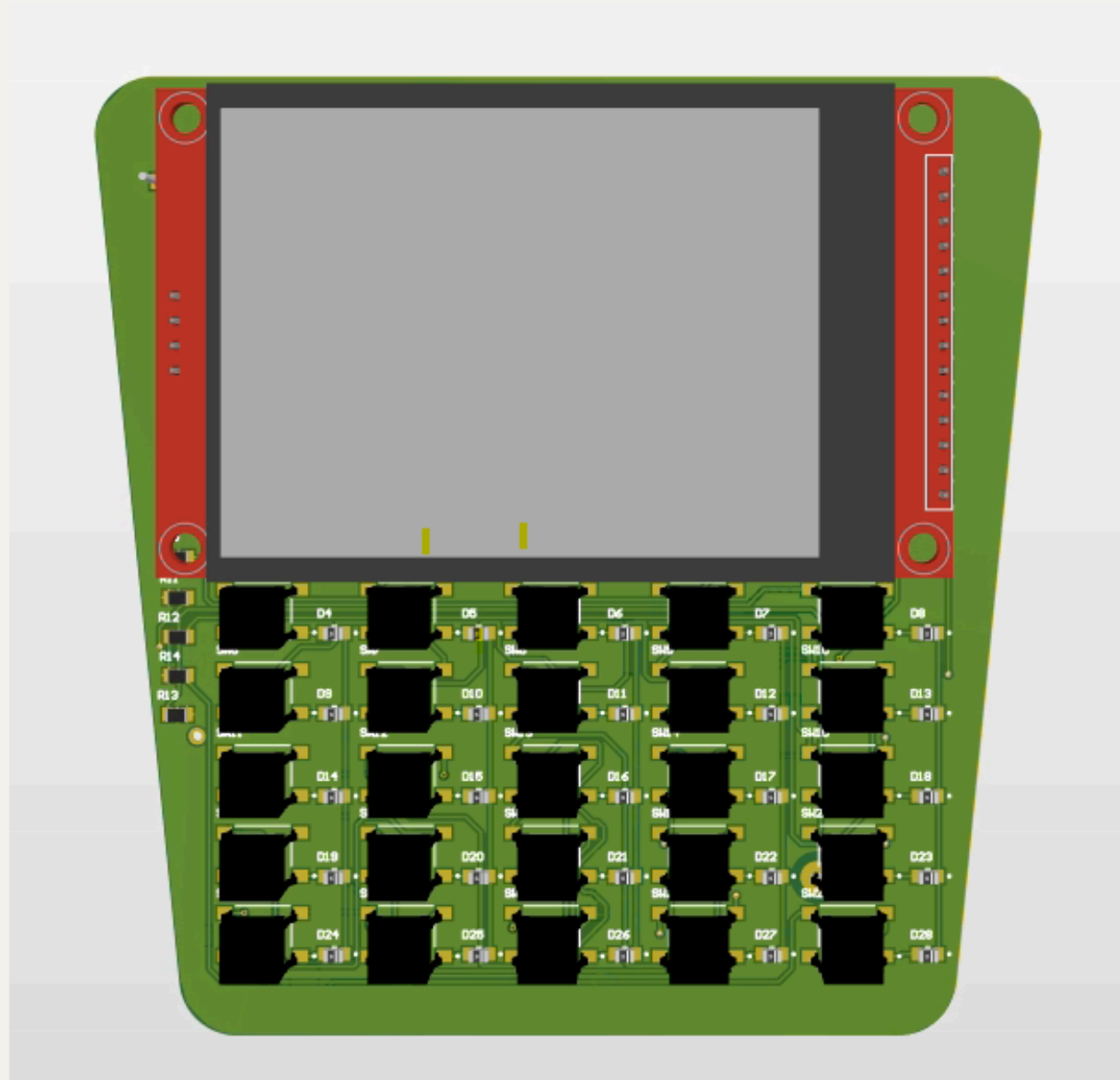
Ticketing device



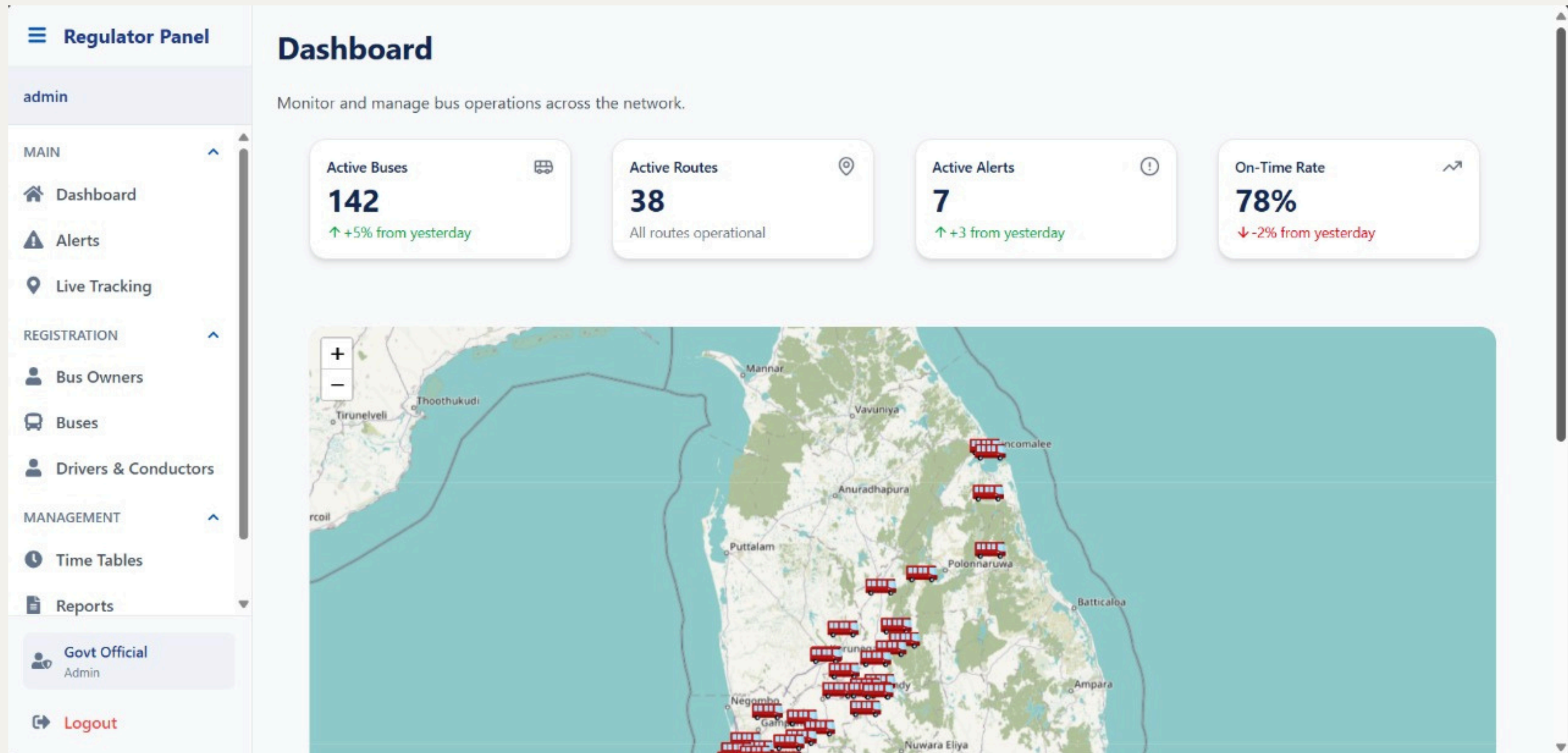
Commuter app



PCB design- iteration 1



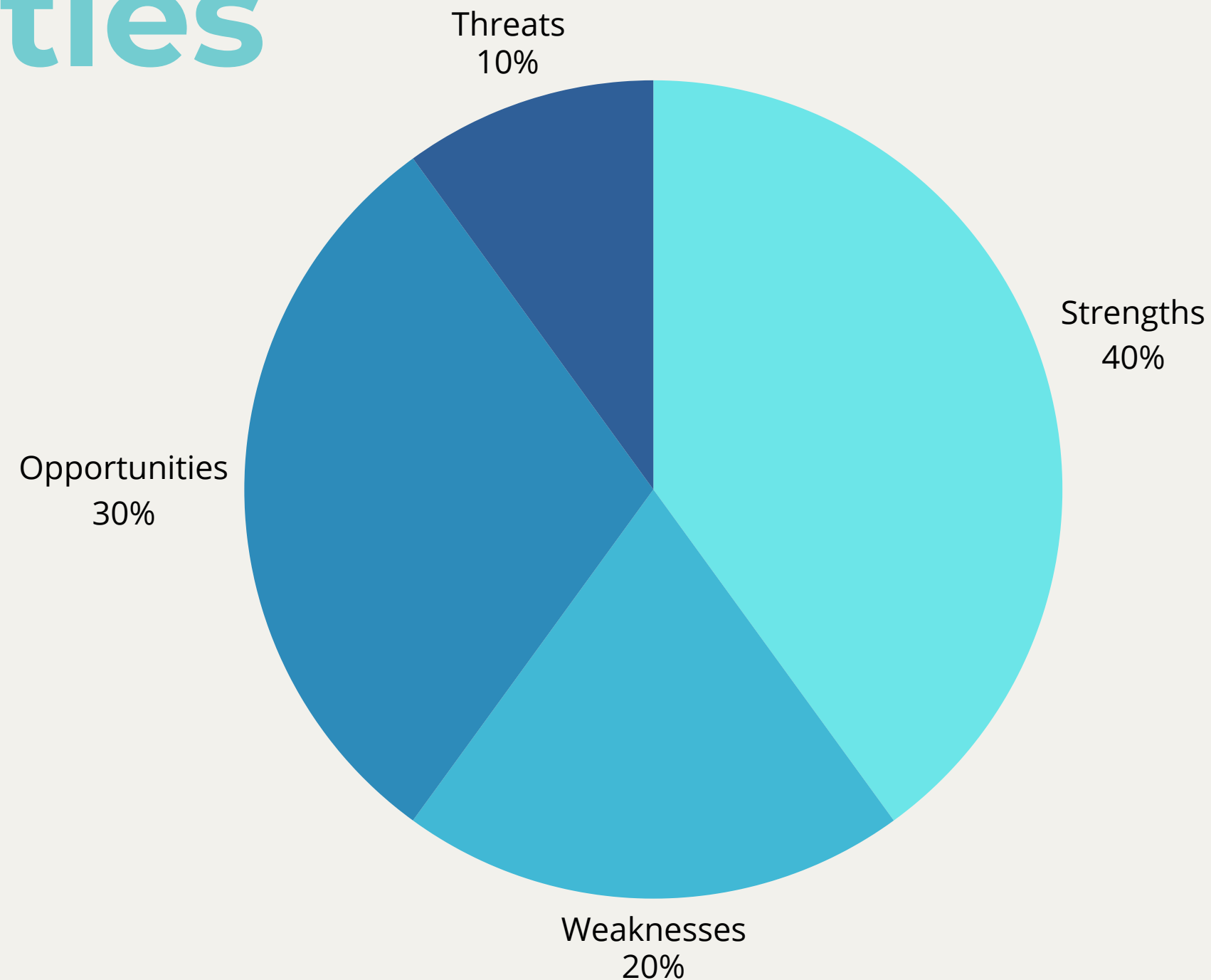
Web portal





Why This System Can Succeed

Strengths, Challenges & Opportunities



Demand & Feasibility

Year	2017	2018	2019	2020	2021	2022
Passengers travelled on Tickets (annual)	749,665,777	717,916,206	647,810,104	396,885,947	279,735,189	530,487,747
Passengers travelled on Season Tickets (annual)	207,038,750	230,674,750	225,926,950	116,955,500	70,769,050	360,392,200
Passengers travelled on Passes(annual)	48,500,856	41,871,340	43,226,950	44,263,308	45,212,550	40,475,580

Ministry of Transport, Highways, Ports and Civil Aviation

- Over **60%** of public transport users struggle with inefficiency & fraud.
- GPS tracking is a must → Countries like India, Singapore, and the UK have already implemented similar systems.
- Sri Lanka has **over 70%** smartphone penetration

The Computer Literacy Statistics (2019) released by the Census and Statistics Department showed that 72.2% of Sri Lankans between the ages of 5-69 years preferred to log onto the internet using smartphones

- Expanding 4G & 5G networks
- Sri Lanka's push for **digital transformation & smart city projects** supports this initiative.



Business Plan & Revenue Model

Government doesn't need to keep funding this. It's self sustaining.

- **Digital Ticketing Fees** → A small service fee per ticket .
- **Advertising Revenue** → Sell ad space on bus stops, ticketing machines, and the app.
- **Data Monetization** → Transport analytics for private companies

How We Make This Happen



No. Buses	500 buses in Colombo	Scale to 3,000+ intercity buses.	Island - Wide Execution
Time period	First 6 months	6 - 18 months	18+ Months
Additional	Passenger app + real-time tracking system goes live.	Introduce AI-driven route optimization.	International partnerships & licensing opportunities.



How We Mitigate Risks

- **Resistance from Private Bus Owners ? - Bring new rules and regulations**
- **High Initial Costs? - Staggered investment plan + revenue generation starts early.**
- **Passenger Adoption? - Public awareness campaigns + incentives for digital payments.**



Thank You!



We are SYNTECH

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